

ELECTIVE COURSE-9.4

URA744: DEEP LEARNING				
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	2	0	2	3.0
Course Objective: To introduce students to the fundamental concepts and techniques of Deep Learning. By the end of the course, students will be able to use Python deep learning libraries to build, train and evaluate basic Neural Networks, Convolutional Neural Networks, and Autoencoders. Additionally, students will gain an understanding of state-of-the-art industry relevant advanced topics in Deep Learning.				
Syllabus Introduction to Deep Learning: Introduction to Deep Learning and Neural Networks, Understanding the structure of Neural Networks and activation functions (e.g. ReLU, sigmoid, tanh etc.), Deep learning libraries. Deep Neural Networks (DNNs): Understanding DNNs and their architecture, Multi-layer perceptron, Building a basic Neural Network, Understanding backpropagation and gradient descent, Tuning hyperparameters for Neural Networks. Convolutional Neural Networks (CNNs): Understanding CNNs and their architecture, understanding convolutional layers and filters, building a basic CNN model, Preprocessing and data augmentation, Training CNNs for classification tasks. Evaluating CNN performance with metrics such as accuracy, precision, and recall. Optimizing CNNs with different optimization algorithms and learning rate schedules. Autoencoders: Understanding Autoencoders and their architecture, Building a basic autoencoder model, Encoding and decoding data, Training and evaluating models, Evaluating the performance of autoencoders using reconstruction error and other metrics. Advanced Topics in Deep Learning: Introduction to advanced topics in deep learning such as regularization and optimization techniques, Understanding the trade-offs between different techniques and selecting the appropriate ones for state-of-the-art industry relevant problems. Research Assignment: Students will be divided into groups to finish course projects. The course projects will consist of actual data from industrial test setups. The course project requires students to use scientific programming and deep learning tools learned from this course to deal with real life problems.				
Laboratory Work (if applicable) 1. Writing Python programs for setting up deep learning libraries such as Tensorflow or PyTorch and getting familiar with the basic functionalities of the libraries.				

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2. Writing Python programs to implementing a basic neural network with a single hidden layer using a deep learning library.
3. Writing Python programs for implementing a convolutional neural network using a deep learning library.
4. Writing Python programs for fine-tuning a pre-trained CNN for a specific classification task.
5. Writing Python programs for implementing an autoencoder using a deep learning library.
6. Writing Python programs for implementing regularization techniques such as L1/L2 regularization and
7. Writing Python programs for implementing advanced topics in deep learning such as optimization techniques like Adam, RMSprop, and learning rate schedules like learning rate decay.

Course Learning Outcomes (CLOs)

The students will be able to:

1. analyze and design deep neural network architectures for a given problem.
2. build and train Convolutional Neural Networks (CNNs) to solve classification/regression problems.
3. apply the basic concepts and principles of autoencoders and their variants.
4. communicate data-driven insights in multiple media mode including graphs, charts, and histograms.

Text Books

1. *Ian Goodfellow and Yoshua Bengio and Aaron Courville, "Deep Learning", MIT Press, 2016.*
2. *Michael Nielsen, "Neural Network and Deep Learning", Online Book 2016.*

Reference Books

1. *Le Deng and Dong Yu, "Deep Learning: Methods and Applications", Foundations and Trends in Signal Processing, 2013.*
2. *Charu C. Aggarwal, "Neural Networks and Deep Learning", Springer; 1st ed. 2018*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessional (Including assignments/ Minor Projects/ Quizes etc.)	35

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